

# **PAPER\_1601 — Pine Wood Density = $\rho_{\text{pine}} = \text{SO}_5^2 \cdot \text{D}_{\text{phys}} + \text{SO}_5^2 = 400 + 100 = 500 \text{ kg/m}^3$**

**Author:** Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Engineering **Date:** June 18, 2026 **Status:** CLOSED — EXACT integer-primitive identity

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## **Observation**

PAPER\_1209Y\_S570 (Engineering Unified Proof Set) lists **Pine Wood Density** with the integer-primitive expression:

$$\rho_{\text{pine}} = \text{SO}_5^2 \cdot \text{D}_{\text{phys}} + \text{SO}_5^2 = 400 + 100 = 500 \text{ kg/m}^3$$

Residual: **EXACT**.

## **UQFF Closed Identity**

$$\rho_{\text{pine}} = \text{SO}_5^2 \cdot \text{D}_{\text{phys}} + \text{SO}_5^2 = 400 + 100 = 500 \text{ kg/m}^3 \quad \text{EXACT}$$

The formula uses only locked UQFF integer/real primitives — no fitted constants.

## **NOT REPLACEMENT**

Empirical engineering measures this constant directly. UQFF supplies a structural derivation from the integer primitives.

## **Reference**

- Source: PAPER\_1209Y\_S570
- Related: PAPER\_1494-1595 (other session-2026-06-18 closures)
- Calculator dispatch: `calculate_paradox({"paradox": "pine_wood_density_500"})`

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